

CTF as a Learning Engine — Run Sheet (30 min, hard stop)

Context: 30 min slot carved out of a 15-min break — audience will feel the clock. No slack behind you. End on time, every time.

Cut-first list (decide now, not on stage):

1. Gender/inclusion slide — first cut
 2. “Dystopian sci-fi” cultural aside — trim to one line or cut
 3. Job-market stat — already folded into opener as one sentence; don’t expand live
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0:00–0:30 | Cold open — the hook

No slide, or just your name/title. Drop straight into: *“In 2019 I’d already spent a decade in information security. I sat down to play my first CTF and asked myself a question I didn’t expect: is this more interesting than my actual job?”* Don’t answer it yet. Let it hang. Move to data.

0:30–3:00 | Why this matters at all (the macro case)

- Global learning poverty stat: 7 in 10 kids in LMICs can’t read a simple paragraph by age 10; 273M children out of school (2024)
- Pivot: education is highest-leverage investment we have — \$196T GDP case for universal foundational learning
- Cybersecurity layered on top: ISC2 — 10.2M jobs demanded vs 5.5M workforce, ~4.76M gap
- Sharper point: 2025 ISC2 study stopped publishing a single gap number — skills shortfalls now matter more than headcount; 95% report a skill need, 88% had a breach tied to skills gap
- One-line job market close: *“This isn’t a nice-to-have — it’s a structural gap, and it’s not closing with headcount alone.”*

- (Optional, cuttable) one line on pop culture casting hackers as protagonists — flag as cultural signal, not prediction, move on fast

Target: hit 3:00, no later.

3:00–6:00 | AI: friend, not enemy — but underused

- AI Kompas survey (Scio Research, led by Miroslava Hapalová, 6,700+ teachers/students, SK+CZ) — verify “Beránová” co-author claim before this slide goes live, couldn’t confirm it
- Core stat: only 26% of teachers use AI directly with students in lessons; most AI use is lesson prep/admin
- A third of teachers think kids should learn *about* AI but not *use* it
- Reframe: hesitancy isn’t fear of risk, it’s lack of training/support (CZ teachers: time 31%, tech support 28%, risk 30%)
- Equity angle: AI tutors help weakest students the most
- Human beat: Hapalová’s daughter — “Mama, I enjoy creating this, I don’t want AI to do it for me” — protecting her own competence
- Bridge line: *“Even AI-literate professionals default to using powerful tools around the task, not inside it. That’s the same instinct that’s kept hands-on learning out of security classrooms for years.”*

Target: hit 6:00.

6:00–9:00 | Your story, part 1 — getting humbled

- 2019, a decade of professional security experience, sat down to play — and wasn’t good at it
- CTF showed you what you didn’t know, despite the resume
- This is your pivot into “why did this hook me instead of discourage me”

9:00–12:30 | Self-Determination Theory — why CTF beat the job

- Deci & Ryan, SDT, 1985/2000 — one of the most cited papers in psychology (80,000+ citations)

- Three needs: autonomy, competence, relatedness
- Map directly onto your story:
 - Autonomy — you choose the challenge, the order, the tools; work doesn't give you that
 - Competence — flag or no flag, immediate unambiguous feedback vs. slow/political work feedback
 - Relatedness — team e0_, the community, BSides stage moment
- The sharper point: it's not extrinsic-bad/intrinsic-good — it's about internalization. Job = compliance. CTF = identity (your own brand, your own writeups site).
- Teacher bridge: SDT needs aren't preferences, they're needs — like food or sleep. A good CTF-style activity isn't a gimmick, it supplies psychological nutrients a lecture often can't.

12:30–14:30 | Flow theory — what it feels like

- Csikszentmihalyi — flow as the challenge/skill balance state
- CTF's tiered difficulty is a near-natural implementation of this
- Honest caveat: pure continuous flow ≠ best retention. Structured interruption — reflection, writeups, debriefs — improves retention over flow alone.
- Teacher payoff: this is *why the teacher's role still matters* even in a game-based format. Scaffolding and reflection aren't anti-fun, they're what make the learning stick.

14:30–18:30 | The dopamine question — is this just gambling?

Keep this tight. Don't let it expand live — it's the section most likely to invite tangents.

- Same neurotransmitter, different system behavior
- Gambling disorder corrupts the *anticipation* signal — reward decoupled from skill, variable-ratio schedule, resistant to extinction
- CTF reward is skill-coupled and effort-coupled — you solve harder challenges because you got better, not because of a random payout
- Tolerance/blunting: addiction literature shows blunted neural reward response over time (ventral striatum, prefrontal cortex) in pathological gambling — no equivalent evidence for puzzle-solving; it's associated with mastery, not numbing

- Honest reframe: the *mechanism* (dopamine) isn't the harm — true of food, exercise, love too. Risk factors are: reward decoupled from effort, no stopping point, escalating stakes. A well-designed CTF doesn't have these structurally. A poorly designed one, or compulsive solo grinding with no off-switch, still could.
- Teaching point: **the form of the reward system matters more than the chemical.**

18:30–23:00 | Your story, part 2 — the cheating incident

This is your strongest material. Give it room. Don't rush.

- You cheated, got to first place, organizer asked you questions, you didn't know the answers, got ditched publicly
- But: you'd also done a huge amount of legitimate work yourself — and you couldn't let it go, because you'd *earned* parts of it honestly
- This is what got you truly hooked — not the win, the discomfort and the refusal to accept an unearned result
- Payoff, tying back to dopamine section: a pathological gambler chasing losses doesn't feel that discomfort, doesn't reflect, just doubles down. Your reaction is a prefrontal-cortex-intact response — proof this isn't compulsion, it's something else.
- Payoff, tying back to SDT: you can steal the rank, you can't steal the competence. The cheat got you points (extrinsic) but none of the mastery (intrinsic) — and that gap is what you actually felt.
- On cheating in CTF generally (brief, ~30 sec): no good prevalence data exists for CTF specifically — say so plainly. But structurally, CTF is harder to get away with cheating in than a written exam, because the "proof of work" is supposed to be reproducible by the person who did it — which is exactly how you got caught.

(If short on time, this is the section to protect — cut elsewhere, not here.)

[CUTTABLE] 23:00–25:00 | Gender & inclusion in CTF learning

- Francis et al., 2-year study using CTF as the intervention
- Self-efficacy gains for both genders — CTF closes confidence gap, not just skills gap
- Motivation style differed: men trended more intrinsic post-training, women more extrinsic — not a deficit, a design signal
- Caveat: literature on gender + cyber-confidence is mixed generally; the gap is often

confidence, not competence — and CTF's objective pass/fail format is unusually good at correcting that

- One-liner if cutting: *"CTF doesn't just close the skills gap — done right, it closes the confidence gap too, for everyone in the room."*

25:00–28:00 | What this means for the room — practical takeaways

- CTF-style learning isn't a gimmick bolted onto teaching — it's a vehicle for needs lecture-based classrooms often can't supply
- But: needs scaffolding, debrief, reflection — flow without structure leaks retention
- Format matters for who it reaches — design the room, not just the difficulty curve
- Practical: doesn't require a real CTF platform — scaffolded, low-stakes, skill-coupled challenge-and-feedback loops are the transferable principle

28:00–30:00 | Close

- Callback to the cold open: *"Is this more interesting than the job? Yes — and now you know exactly why, mechanistically. Not because play is silly and work is serious. Because one of them was built around what brains actually need, and the other usually isn't built at all."*
- Hard stop. Do not start Q&A inside this slot — let the room have its break, take questions after if people stay.

Pacing notes

- Rehearse with a visible timer, not a mental estimate — the cheating story will run long if you let it breathe, which you should let it do, but compensate elsewhere
- If you're past 18:30 minute-mark before starting the cheating story, cut the gender section preemptively rather than rushing the story
- If you're past 21:00 before starting the cheating story, also trim the dopamine section's tolerance/blunting detail live — keep only the skill-coupled vs. variable-ratio distinction
- Protect: cold open, cheating story, SDT mapping, final callback. Everything else is negotiable under time pressure.

